Xi Lin

lx20001209@gmail.com github.com/EricLin09 ericlin09.vercel.app Sydney, Australia

Summary

Mathematics graduate (Imperial College London) and IT postgraduate (UTS) building toward a Machine Learning Engineer role. Strong foundation in linear algebra, probability, and numerical methods, paired with hands-on experience implementing ML and deep learning systems in Python. Recently shipped an end-to-end LLM application from prototype to a deployed public demo. Seeking a machine learning / data internship for Summer 2026–27.

Skills

Programming: Python, R, SQL
ML & Deep Learning: Deep learning, LLM application development (RAG, embeddings, vector search), clustering, statistical learning
Frameworks & Tools: PyTorch, scikit-learn, LangChain, ChromaDB, FastAPI, NumPy, Pandas, OpenAI API, Git/GitHub, AWS Cloud
Mathematics: Linear Algebra, Probability & Statistics, Optimization, Time Series Analysis
Languages: Chinese (native), English (fluent)

Projects

StudyMate — RAG Study Agent with Live Public Demo

Personal Project — GitHub — Live Demo *2026*

- Built a RAG agent over lecture notes (Python, LangChain, ChromaDB, OpenAI API) with two modes: grounded Q&A that cites source file and page for every answer, and an auto-quiz mode that generates short-answer questions from the knowledge base and grades responses with LLM feedback.
- Built an automated retrieval evaluation pipeline: LLM-generated test set with ground-truth chunks, benchmarking dense, BM25, and RRF-hybrid retrieval on Hit@4 / MRR@10; dense retrieval achieved **96% Hit@4** (MRR 0.917) on a 700-chunk course corpus, with error analysis of why equal-weight hybrid fusion underperformed.
- Designed an incremental indexing pipeline using SHA-256 content fingerprints, re-embedding only new or modified files and pruning deleted ones from the vector store, cutting redundant embedding cost to zero on re-ingestion.
- Deployed as a public, rate-limited demo embedded in my portfolio site: FastAPI backend with per-visitor isolated vector sessions (2h TTL, nothing persisted), streaming responses, per-IP and daily request caps, and HMAC-signed grading requests to prevent abuse as a generic LLM proxy.

Loan Default Prediction — Credit Risk Classification

Personal Project — GitHub — Write-up *2026*

- Built a leakage-free ML pipeline on 18K loan records to predict borrower default, benchmarking 9 classifiers and a soft-voting ensemble with 5-fold stratified cross-validation and `GridSearchCV` tuning.
- Benchmarked imbalance strategies (baseline vs. class weights vs. SMOTE) on identical CV folds and showed neither beat the baseline on minority-class F1 — published the analysis; the final gradient-boosted model reached **ROC-AUC 0.89** and **0.94 precision**.
- Interpreted predictions via SHAP and permutation importance, surfacing employment status, credit score, and debt-to-income ratio as the top default drivers.

Experience

National Supercomputing Center Wuxi Wuxi, China

Research Intern, Intelligent Computing Department *Oct 2024 – Jan 2025*

- Helped build a patent-valuation system based on textual semantic similarity, estimating a patent's value from its semantic relatedness to the prior decade of patents (higher relatedness indicating higher value).
- Implemented word2vec embeddings in PyTorch to represent patent text, providing the semantic vector foundation for downstream similarity computation.

- Configured and ran model training on a remote GPU / Linux environment, handling training execution and job scheduling.
- Contributed to a separate research project applying Google AlphaFold3 to protein-structure prediction.

Education

University of Technology Sydney — Sydney, Australia
Master of Information Technology — *Jul 2025 – Jun 2027 (expected)*

Imperial College London — London, UK
B.Sc. in Mathematics — *Sep 2020 – Jun 2024*

Competitions & Honors

- Dean's List, University of Technology Sydney, Autumn 2026.
- Mathematical Contest in Modeling (MCM), 2021 — Meritorious Award (team).
- AMC Mathematics Competition, 2019 — Gold Award.